# LLM-Based Summarization of Fragmented ICU Clinical Notes: A Multi-Model, Multi-Evaluator Study of Zero-Shot and Retrieval-Augmented Generation

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## Abstract

Clinical information in ICU electronic health records is fragmented across multiple note types, requiring physicians to manually synthesize large volumes of documentation during patient handoffs and care transitions. We investigate whether large language models can generate accurate Brief Hospital Course (BHC) summaries from multi-source ICU notes, and whether retrieval-augmented generation (RAG) improves on naive zero-shot prompting. Using the VeriFact-BHC corpus (100 MIMIC-III patients, 4,787 clinical notes, 13,070 clinician-annotated propositions), we compare zero-shot and budget-matched RAG pipelines across two open-source model families (Mistral 7B Instruct and Llama 3.1 8B Instruct) and evaluate outputs with three complementary methods: reference-based metrics (ROUGE, BERTScore), proposition-level fact recall against clinician-verified ground truth, and an LLM-as-judge evaluation using Qwen3-32B with a five-dimension clinical rubric. We additionally replicate the comparison on a curated single-document corpus (MIMIC-IV-Ext-BHC) as a cross-dataset ablation. Reference-based metrics show that RAG matches zero-shot performance (ROUGE-L 0.168 vs. 0.165 for Mistral; 0.159 vs. 0.159 for Llama) while consuming roughly 42% less input context; proposition recall is similar across conditions ($\sim$73%); and LLM-as-judge scores favor RAG on completeness and overall quality, while baseline retains a small edge in conciseness. Hallucination scores are uniformly compressed across conditions despite a qualitative anchor case demonstrating clinically significant hallucination differences. Together, the multi-method evaluation reveals that no single metric captures BHC quality completely, that RAG's primary benefit at this scale is computational efficiency rather than higher quality scores, and that retrieval is most valuable precisely when the source documentation is fragmented enough to require it.

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## 1 Introduction

Electronic health records contain rich clinical information, but in intensive care units that information is scattered across nursing notes, physician progress notes, consult notes, radiology reports, laboratory results, and procedure documentation. Clinicians transitioning care of a patient, during handoffs, sign-outs, or admission to a new service, must manually synthesize dozens of documents to reconstruct the clinical trajectory. This synthesis is time-consuming, cognitively demanding, and a recognized source of communication error.

Recent work has shown that large language models can produce clinical summaries comparable in quality to those written by physicians [1, 2]. However, much of the prior work focuses on summarization of a single, already-curated source document, leaving open the question of how well LLMs handle the multi-source, fragmented documentation characteristic of real ICU stays. Two related questions follow. First, can a general-purpose open-source LLM produce a clinically faithful Brief Hospital Course (BHC) when given the full set of ICU notes for a patient? Second, does retrieval-augmented generation (RAG) improve over naive prompting in this setting, where source documentation is highly fragmented and frequently exceeds the model's context window?

We address these questions on the VeriFact-BHC corpus [3](a 100-patient subset of MIMIC-III [4] with clinician-annotated factual propositions) using two open-source model families and a deliberately multi-method evaluation. Our contributions are as follows:

- We implement and evaluate a zero-shot baseline and a budget-matched multi-query RAG pipeline on the same 100-patient ICU cohort, across two model families (Mistral 7B Instruct, Llama 3.1 8B Instruct), yielding a $2\times2$ comparison of generation strategy and model.
- We evaluate outputs with three complementary methods: reference-based metrics (ROUGE, BERTScore), proposition-level fact recall against the VeriFact clinician-verified ground truth, and an exploratory LLM-as-judge analysis using Qwen3-32B on a five-dimension clinical rubric.
- We replicate the baseline-vs-RAG comparison on a curated single-document corpus (MIMIC-IV-Ext-BHC) as a cross-dataset ablation, isolating the effect of source-documentation structure on the value of retrieval.
- We document both quantitative findings and a representative qualitative case (Patient 72940) that surfaces clinically significant differences between conditions which the automated metrics do not capture.

## 2 Background

**LLMs for clinical text summarization.** Van Veen et al. [1] showed that domain-adapted large language models can match or exceed physician performance on clinical summarization across multiple text types, including radiology reports, progress notes, and discharge summaries. Williams et al. [2] subsequently reported no significant quality difference between LLM-generated and physician-written hospital discharge summaries on a blinded clinician evaluation. Together these findings suggest that off-the-shelf LLMs have reached a level of clinical text generation capability where deployment is plausible, and where evaluation methodology, rather than raw model capability, becomes the primary bottleneck.

**Retrieval-augmented generation for clinical NLP.** For longer source documents that exceed the model's context window, RAG has emerged as a practical strategy to provide the model with only the most relevant content. Alkhalaf et al. [5] reported that combining retrieval with zero-shot prompting improved accuracy of EHR-based summarization when measured by manual clinical verification, while noting that hallucinated content remained an unresolved limitation. The recurring observation across this literature is that improvements in clinical accuracy are often invisible to surface-level text-similarity metrics.

**Evaluating clinical summaries.** Evaluation of clinical text generation is an active area of methodological work. The Provider Documentation Summarization Quality Instrument (PDSQI-9) [6] was developed and validated specifically for scoring LLM-generated clinical summaries across nine attributes including accuracy, completeness, organization, and clinical utility. A companion study [7] demonstrated that an LLM-as-judge using a reasoning model can achieve substantial agreement with physician evaluators on the PDSQI-9, offering a scalable alternative to manual review. Complementing these instrument-based approaches, the VeriFact framework [3] introduced proposition-level evaluation: decomposing a generated summary into individual clinical claims and verifying each against the source EHR, providing a granular factual-accuracy signal not captured by string-matching metrics.

# 3 Data

## 3.1 Primary Dataset: VeriFact-BHC

Our primary dataset is the MIMIC-III-Ext-VeriFact-BHC corpus [3], a curated cohort of 100 MIMIC-III ICU patients designed for evaluating LLM-generated clinical text. For each patient, the dataset provides all clinical notes from the ICU stay, a human-written Brief Hospital Course extracted from the discharge summary, and a set of clinician-annotated propositions with ground-truth verification labels (Supported, Not Supported, or Not Addressed relative to the source EHR). Table 1 summarizes the cohort characteristics.

Table 1: VeriFact-BHC dataset characteristics (100 MIMIC-III ICU patients). Token counts are computed with the Mistral 7B tokenizer on raw concatenated note text; truncation counts in subsequent tables include the full prompt template wrapped around the notes, which raises the effective truncation rate from 31% to 42% under the Mistral pipeline.

| Characteristic | Value |
|---|---|
| Patients | 100 |
| Hospital admissions | 125 (25 patients with prior admissions) |
| Total clinical notes | 4,787 |
| Note categories | 14 (Nursing, Physician, Radiology, ECG, etc.) |
| Notes per patient | Median 25 (range 10–314) |
| Tokens per patient | Median 17,562 (range 5,720–232,852) |
| Patients with raw note tokens > 32K | 31/100 |
| Human BHC length | Median 513 tokens (range 107–1,905) |
| Total propositions | 13,070 |
| Human-authored Supported propositions (recall checklist) | 5,396 |

We chose MIMIC-III over MIMIC-IV for the primary analysis because MIMIC-III contains 14 categories of clinical notes (nursing notes, physician progress notes, consult notes, radiology reports, and others), whereas MIMIC-IV-Note contains only discharge summaries and radiology reports. The diversity of note types in MIMIC-III is essential for evaluating retrieval in a realistic multi-source setting.

A key dataset characteristic that shapes the interpretation of all downstream results is that 25.6% of propositions in the human-written BHCs are labeled “Not Addressed” in the MIMIC-III record, because clinicians who wrote the original summaries had access to information beyond what is captured in the EHR (e.g., bedside conversations, verbal handoffs). Any model limited to MIMIC-III notes therefore cannot be expected to reproduce all content in the human reference, and moderate ROUGE scores against the human BHC should be interpreted as approaching an empirical ceiling rather than reflecting summarization failure.

Preprocessing involved tokenizing all notes using the Mistral tokenizer, computing per-note and per-patient token counts, and saving processed data as Parquet files for efficient loading. All notes were retained without filtering by category to preserve the full information landscape available to the retrieval pipeline.

## 3.2 Ablation Dataset: MIMIC-IV-Ext-BHC

Our secondary dataset is the MIMIC-IV-Ext-BHC corpus [8], used as a controlled ablation setting in which retrieval is largely unnecessary. The dataset contains 270,033 ICU patient records and pairs the non-BHC sections of each discharge summary with the corresponding human-written Brief Hospital Course. Critically, the input is already pre-curated as a single coherent document, in contrast to the fragmented multi-source notes in VeriFact-BHC.

Table 2: MIMIC-IV-Ext-BHC dataset characteristics (curated single-document setting).

| Characteristic | Value |
|---|---|
| Patients | 270,033 |
| Median input tokens | 2,166 |
| Input token range | 67–13,552 |
| Patients exceeding 32K tokens | 0/270,033 |
| Median BHC length | 453 tokens |
| BHC token range | 1–6,341 |

Compared to VeriFact-BHC, the MIMIC-IV-Ext-BHC inputs are an order of magnitude shorter and never exceed the 32K context window. This makes the dataset a useful counterfactual: if RAG's value derives from intelligent selection within an oversized, heterogeneous input, then a setting where the input is already short and curated should attenuate or eliminate that value. We use this dataset for an ablation comparison in Section 5.6.

## 4 Methods

We compare a zero-shot baseline and a retrieval-augmented generation pipeline across two model families on the VeriFact-BHC cohort, and replicate the comparison on the MIMIC-IV-Ext-BHC ablation dataset. All generated BHCs are evaluated with three complementary methods: reference-based metrics, proposition-level fact recall, and an exploratory LLM-as-judge analysis. We describe each component in turn.

### 4.1 Zero-Shot Baseline

The baseline follows the most common configuration in clinical summarization research: zero-shot prompting with a general-purpose instruction-tuned LLM [1]. For each patient, all clinical notes from the ICU stay are concatenated in chronological order and provided to the model with a single instruction to generate a Brief Hospital Course summary. No retrieval, structured output template, or in-context examples are used.

We use Mistral 7B Instruct v0.3 [9] as the primary model, loaded in 4-bit quantization ($\sim$4.1 GB VRAM) on a Google Colab A100 GPU. The model has a 32K-token context window. For patients whose concatenated notes plus prompt template exceed this limit (42 of 100 patients), we truncate from the end, retaining the earliest notes in the sequence; this preserves the admission note, initial workup, and early assessments that form the backbone of most BHC narratives. Generation parameters are temperature 0.1, top-$p$ 0.9, repetition penalty 1.1, and 1,024 maximum new tokens—conservative settings that favor deterministic, factual output appropriate for clinical text. The same generation parameters are used across all conditions and both models so that quality differences reflect input selection rather than decoding stochasticity.

### 4.2 Retrieval-Augmented Generation Pipeline

The RAG pipeline replaces naive note concatenation with targeted retrieval of relevant clinical content within a controlled token budget.

**Chunking.** All 4,787 clinical notes are split into 23,663 chunks of approximately 300 tokens each, with 50-token overlap between consecutive chunks to preserve context across boundaries. Chunking is performed using the Mistral tokenizer to ensure alignment with the generation model's vocabulary; the resulting chunk byte content and per-chunk metadata are reused identically across both model families.

**Embedding.** Each chunk is embedded into a 1,024-dimensional vector using BGE-M3 [10], the same embedding model used in the VeriFact framework [3]. Embedding all chunks takes approximately 221 seconds on a single A100.

**Indexing.** We build one FAISS [11] index per patient using cosine similarity. Per-patient indexing scopes retrieval to a single patient's documentation and prevents cross-patient information leakage. Indexing 100 patients ($\sim$24K chunks) completes in under one minute on FAISS-CPU.

**Retrieval.** We tested three retrieval strategies iteratively to isolate design choices:

- **RAG v1** (single query): A single generic query ("hospital course summary") retrieving the top-40 chunks per patient, yielding $\sim$11K tokens of context.
- **RAG v2** (multi-query): Six clinically motivated queries targeting distinct BHC components (admission and presentation, diagnoses and problems, diagnostic workup, treatment and medications, clinical course and complications, and discharge status) with deduplication across queries, yielding $\sim$10K tokens.
- **RAG v3** (multi-query, budget-matched): The same six clinical queries, with the per-patient token budget raised to match the baseline's effective context window ($\sim$28K tokens), ensuring a fair comparison between baseline and RAG.

The progression from v1 to v3 was driven by two findings during development. First, single-query retrieval disproportionately returned templated content (medication reconciliation lists, ICU order sets) rather than clinical narrative, motivating the switch to multi-query retrieval with clinically structured queries. Second, we identified that v1 and v2 fed the model roughly one-third the context that the baseline used ($\sim$10K vs. $\sim$31K tokens), creating an unfair comparison. RAG v3 corrected this by matching the token budget. We treat v3 as the primary RAG condition for cross-model comparison and judge evaluation.

**Generation.** Retrieved chunks are reordered chronologically by source-note timestamp and passed to the generation model with the same prompt template and decoding parameters as the baseline. The only difference between baseline and RAG is the input content: full concatenated notes (with end-truncation if necessary) versus a budget-bounded retrieved selection.

### 4.3 Cross-Model Comparison: Adding Llama 3.1 8B Instruct

To assess whether findings generalize beyond a single model family, we ported both pipelines to Llama 3.1 8B Instruct [**?** ]. The port preserves the BGE-M3 embeddings, FAISS indices, chunk content, retrieval logic, generation parameters, and per-patient output schema; only the generation model is swapped. Two model-specific adaptations were required. First, we use the Llama chat template (`tokenizer.apply_chat_template`) to produce the correct `<|begin_of_text|><|start_header_id|>` formatting in place of Mistral's `[INST]...[/INST]`. Second, the truncation budget is computed against the Llama tokenizer, which yields a different per-patient truncation set (34/100 truncated under Llama vs. 42/100 under Mistral) despite identical underlying note content. The Mistral-tokenizer chunk boundaries are retained so that FAISS indices remain byte-identical between pipelines, while the Llama tokenizer governs budgeting and decoding. This 2$\times$2 design ($\{$Mistral, Llama$\}$ $\times$ $\{$Zero-Shot, RAG v3$\}$) yields four BHC parquet files of 100 patients each.

### 4.4 Reference-Based Evaluation

We evaluate generated BHCs against the human-written reference BHC using ROUGE [12] (F1 of unigram, bigram, and longest-common-subsequence overlap) and BERTScore [13] (semantic similarity from contextual embeddings, F1 with the default `roberta-large` backbone). These are the dominant automated metrics in the clinical summarization literature and provide a direct comparison to prior work; we report them while noting in Section 6 their well-documented insensitivity to clinically meaningful changes.

**Proposition-level fact recall.** As a complementary, more clinically grounded automated metric, we leveraged the 5,396 propositions extracted from the human-written reference BHCs that were clinician-verified as Supported by the source EHR. For each (patient, proposition) pair, we compute the cosine similarity (using BGE-M3 embeddings) between the proposition and the generated BHC, counting a fact as "recalled" if the similarity exceeds 0.60. The threshold and embedding model match those used by the VeriFact authors, so recall scores are directly comparable to that framework. This metric measures factual coverage independent of surface wording.

### 4.5 LLM-as-Judge Evaluation

To complement reference-based metrics with attribute-level qualitative scoring, we performed an exploratory LLM-as-judge evaluation. We adopted the conceptual framework of PDSQI-9 [6] and the LLM-as-judge configuration of Croxford et al. [7]—which has shown substantial agreement with physician evaluators on similar clinical text—but used a custom five-dimension rubric (rather than the full nine-attribute PDSQI-9 instrument) for tractability under our compute and timeline constraints. We treat the resulting scores as exploratory and complementary to the reference-based metrics, not as a substitute for either physician review or full PDSQI-9 application.

**Judge model and configuration.** We used Qwen3-32B [**?** ] loaded in 4-bit quantization on a single Colab A100, with thinking-mode disabled and deterministic decoding (`do_sample=False`, max 500 new tokens). The judge was prompted in a strict-JSON-output configuration with an explicit calibration instruction that "most summaries should receive scores of 2–4" on a 1–5 scale, and an evidence hierarchy designating the clinical context as the primary source of truth and the human reference as a secondary anchor for identifying missing events.

**Inputs to the judge.** For each evaluation, the judge received: (i) up to ∼24,000 characters of clinical context constructed from the patient's notes (sorted chronologically; up to 40 notes; up to 1,200 characters per note); (ii) up to 4,000 characters of the human-written reference BHC; and (iii) the full generated summary under evaluation. The judge produced a single JSON object containing five integer scores (factuality, hallucination, completeness, conciseness, overall quality, all on 1–5 scales) plus three structured lists (*major_supported_points*, *major_unsupported_or_hallucinated_points*, *major_missing_points*) and a brief free-text rationale. By rubric convention, factuality, completeness, conciseness, and overall quality are higher-is-better; hallucination is lower-is-better.

**Conditions evaluated.** The judge was applied to all four Mistral conditions (zero-shot baseline, RAG v1, RAG v2, RAG v3) on all 100 patients, yielding 400 single-summary evaluations. Llama outputs were not scored by the judge in this iteration; we discuss this asymmetry as a limitation in Section 6.

**Robustness handling.** A two-stage parser handled JSON extraction with an automatic "repair" fallback that re-prompts the judge to convert non-conforming output into the schema. Across the 400 evaluations, parse-error rates ranged from 1% (RAG v3) to 5% (baseline and RAG v1); these patient–system pairs are excluded from the per-condition score means rather than imputed.

## 5 Results

### 5.1 Reference-Based Metrics

Table 3 reports the Mistral RAG iteration on the full 100-patient cohort. The baseline achieves ROUGE-1 0.339, ROUGE-L 0.165, and BERTScore F1 0.819. RAG v1 and v2 each operate at roughly one-third the baseline's input budget and underperform on ROUGE-L by 1–3 thousandths. RAG v3, with budget matched to the baseline, is the first variant to match or marginally exceed baseline on ROUGE-L (0.168) and BERTScore (0.820), while still using approximately 42% less context on average (∼18K vs. ∼31K tokens).

Table 4 presents the cross-model 2×2 comparison between Mistral and Llama on the same cohort, restricted to the two metrics most directly comparable across pipeline runs. Two patterns are notable. First, both models show the same flat reference-metric profile: zero-shot and RAG v3 score within 0.001 BERTScore of each other for both Mistral and Llama. Second, Llama scores

Table 3: Reference-based metrics for Mistral 7B Instruct: zero-shot baseline and three RAG variants on the 100-patient VeriFact-BHC cohort.

| Metric | Baseline | RAG v1 (single-Q, 11K) | RAG v2 (multi-Q, 10K) | RAG v3 (multi-Q, 28K) |
|---|---|---|---|---|
| ROUGE-1 F1 | 0.339 | 0.339 | 0.335 | 0.334 |
| ROUGE-2 F1 | 0.084 | 0.079 | 0.081 | 0.084 |
| ROUGE-L F1 | 0.165 | 0.163 | 0.162 | **0.168** |
| BERTScore F1 | 0.819 | 0.818 | 0.818 | **0.820** |
| Avg. input tokens | $\sim$31K | $\sim$11K | $\sim$10K | $\sim$18K |

systematically below Mistral on both ROUGE-L (0.159 vs. 0.165–0.168) and BERTScore (0.813–0.814 vs. 0.819–0.820), despite identical retrieval inputs in the RAG conditions. The cross-model gap (5–7 thousandths on ROUGE-L) is larger than the within-model zero-shot-vs-RAG gap (3 thousandths or less), suggesting that at this scale and on this metric profile, the choice of generation model affects scores more than the choice of input strategy.

Table 4: Cross-model reference-based metrics on the VeriFact-BHC cohort (n=100). Truncation counts are model-specific because the prompt template plus notes interact with each model's tokenizer differently.

| | Mistral 7B Instruct | | Llama 3.1 8B Instruct | |
|---|---|---|---|---|
| Metric | Zero-shot | RAG v3 | Zero-shot | RAG v3 |
| ROUGE-L F1 | 0.165 | **0.168** | 0.159 | 0.159 |
| BERTScore F1 | 0.819 | **0.820** | 0.813 | **0.814** |
| Patients truncated | 42/100 | — | 34/100 | — |
| Avg. input tokens | $\sim$31K | $\sim$18K | $\sim$31K | $\sim$18K |

### 5.2 Truncation Subgroup Analysis

Truncation in the zero-shot condition has a measurable, model-consistent effect on output quality. In the Mistral baseline, the 42 patients requiring truncation score lower across all metrics than the 58 patients whose notes fit (ROUGE-L 0.148 vs. 0.177; BERTScore 0.814 vs. 0.823). These patients have a mean of 105,563 raw note tokens versus 17,719 for non-truncated patients, meaning the baseline discards roughly two-thirds of their clinical documentation. The same pattern holds for Llama: in the L-ZS condition, the 34 truncated patients (mean $\sim$100K input tokens) score ROUGE-L 0.141 versus 0.168 for the 66 non-truncated patients (mean $\sim$16K input tokens)—a 0.027 gap that quantifies the cost of positional truncation when context is exhausted (Table 5).

Table 5: Zero-shot truncation subgroup analysis: ROUGE-L F1 by truncation status, by model. Truncation populations are model-specific (see Table 4).

| | ROUGE-L F1 | | |
|---|---|---|---|
| Model (Zero-Shot) | Truncated | Not truncated | Gap |
| Mistral 7B | 0.148 (n=42) | 0.177 (n=58) | 0.029 |
| Llama 3.1 8B | 0.141 (n=34) | 0.168 (n=66) | 0.027 |

This finding directly motivates the RAG pipeline: if intelligent retrieval can substitute for brute-force concatenation, the quality penalty observed for high-token patients should narrow under RAG. Aggregate metrics in Table 4 confirm that RAG v3 recovers to baseline performance overall while consuming substantially less input.

### 5.3 Proposition-Level Fact Recall

Table 6 reports proposition-level fact recall for the Mistral conditions, computed against the 5,396 human-authored, clinician-verified Supported propositions. Both baseline and RAG v3 capture approximately 73% of verified clinical facts, with RAG v3 trailing baseline by 1.0 percentage point overall. The pattern is consistent within both truncation subgroups: baseline retains a small advantage on the recall measure regardless of input regime. We interpret this null result as concordant with the flat reference-metric pattern in Table 3: when retrieval is configured to fairly match the baseline's input budget, no automated metric in our suite detects a systematic accuracy difference between the two strategies.

Table 6: Proposition-level fact recall using VeriFact clinician-verified Supported propositions (cosine similarity threshold = 0.60, BGE-M3 embeddings). Mistral conditions, n=100 patients.

| | Baseline | RAG v3 | $\Delta$ |
|---|---|---|---|
| Mean fact recall | 73.5% | 72.5% | −1.0% |
| Total facts captured | 3,735 / 5,396 | 3,711 / 5,396 | −24 |
| Recall (truncated, n=42) | 70.0% | 69.8% | −0.3% |
| Recall (not truncated, n=58) | 76.0% | 74.5% | −1.5% |

### 5.4 LLM-as-Judge Scores

Table 7 reports the LLM-as-judge scores across the four Mistral conditions. Two patterns are visible. First, the rubric does discriminate among conditions on three of the five attributes: completeness rises monotonically across the RAG progression (from 2.54 in v1 to 2.80 in v3), overall quality follows the same trajectory (2.54 to 2.80), and conciseness is highest for the baseline (3.39) and roughly equivalent across the RAG variants (3.12–3.18). Second, hallucination scores are essentially flat across all four conditions (means 2.95–2.99, with standard deviations as low as 0.10 for the baseline), and factuality scores show only a modest non-monotone pattern across RAG iterations.

Table 7: LLM-as-judge scores (Qwen3-32B, single-summary scoring) on the four Mistral conditions. Mean $\pm$ standard deviation across patients with valid (non-parse-error) judge output. Score range: 1–5; factuality, completeness, conciseness, and overall quality are higher-is-better; hallucination is lower-is-better.

| Condition | Score (mean $\pm$ SD) | | | | |
|---|---|---|---|---|---|
| | Factuality ($\uparrow$) | Hallucination ($\downarrow$) | Completeness ($\uparrow$) | Conciseness ($\uparrow$) | Overall ($\uparrow$) |
| Baseline (n=95) | $2.92 \pm 0.31$ | $2.99 \pm 0.10$ | $2.64 \pm 0.48$ | $\mathbf{3.39 \pm 0.49}$ | $2.64 \pm 0.48$ |
| RAG v1 (n=95) | $2.69 \pm 0.46$ | $2.95 \pm 0.22$ | $2.54 \pm 0.50$ | $3.12 \pm 0.32$ | $2.54 \pm 0.50$ |
| RAG v2 (n=97) | $2.80 \pm 0.40$ | $2.96 \pm 0.20$ | $2.65 \pm 0.48$ | $3.18 \pm 0.38$ | $2.65 \pm 0.48$ |
| RAG v3 (n=99) | $\mathbf{2.92 \pm 0.34}$ | $2.97 \pm 0.17$ | $\mathbf{2.80 \pm 0.40}$ | $3.16 \pm 0.37$ | $\mathbf{2.80 \pm 0.40}$ |

We note that the per-condition mean for *overall_quality* is numerically identical to that for *completeness* in every row of Table 7; the judge appears to anchor its overall verdict tightly to its completeness assessment, a behavior that further supports treating these scores as exploratory rather than as five fully independent quality dimensions. The directional finding—that RAG v3 modestly improves on baseline in the dimensions retrieval was designed to help (completeness and overall quality) while baseline retains a small edge in conciseness—is consistent with the qualitative case study in Section 5.5, in which RAG v3 identifies surgical detail that the baseline misses. The flat hallucination distribution is at first surprising but, as we discuss in Section 6, plausibly reflects (a) the calibration instruction compressing scores into the middle of the 1–5 scale, (b) limitations of a single judge model in detecting hallucination at this output length, or (c) genuine equivalence in hallucination rates across the four Mistral conditions. We do not attempt to discriminate among these explanations without physician calibration, and treat the judge results as one of three complementary evaluation signals rather than as a definitive ranking.

### 5.5 Qualitative Anchor Case: Patient 72940

While the automated metrics show approximate equivalence, qualitative clinical review surfaces differences invisible to ROUGE, BERTScore, and proposition-level recall. Patient 72940 (a cardiac surgery admission with $\sim$37K tokens of notes) is illustrative: the zero-shot baseline produced a summary stating an admission diagnosis of "aortoiliac disease," which is not supported by the source notes. The RAG v3 pipeline, in contrast, correctly identified the surgical procedure with specific graft details (LIMA-LAD, SVG-OM) that match the human-written reference BHC. The corresponding ROUGE-L improvement on this patient was substantial ($0.113 \rightarrow 0.240$), but the clinical significance of replacing a fabricated diagnosis with the correct one goes well beyond what the metric increment conveys.

In the opposite direction, Patient 61157 (subarachnoid hemorrhage, $\sim$86K tokens) illustrates a failure mode of RAG: the baseline produced a more coherent narrative from chronologically ordered early notes, while the RAG output was longer but less focused. These two cases bracket the qualitative range and reinforce the broader observation that aggregate ROUGE/BERTScore parity in Table 4 hides per-patient swings in clinically relevant content—a phenomenon also documented by Alkhalaf et al. [5], who found that the value of RAG for clinical summarization was visible only under manual clinical verification.

### 5.6 Cross-Dataset Ablation: Curated vs. Fragmented Input

To assess whether the value of retrieval depends on the structure of the source documentation, we replicated the baseline-vs-RAG comparison on the MIMIC-IV-Ext-BHC dataset (n=100) using Flan-T5-Large, a 770M-parameter encoder-decoder model with a 512-token input budget. Retrieval used S-PubMedBert-MS-MARCO embeddings with character-level chunking (400 characters, 80-character overlap) and top-3 chunk retrieval per patient.

Table 8: Ablation results on MIMIC-IV-Ext-BHC: zero-shot baseline vs. RAG (Flan-T5-Large, n=100). In contrast to the VeriFact-BHC findings (Tables 3, 4), RAG underperforms baseline on every metric in the curated-input setting.

| Metric | Baseline | RAG |
|---|---|---|
| ROUGE-1 F1 | **0.1355** | 0.0941 |
| ROUGE-2 F1 | **0.0444** | 0.0353 |
| ROUGE-L F1 | **0.0921** | 0.0688 |
| BERTScore F1 | **0.7471** | 0.7291 |

The pattern is opposite to that observed on VeriFact-BHC: on the curated single-document dataset, RAG underperforms baseline on every metric. We attribute this to the structure of the input. When the source document is already compact and fits entirely within the context window, selecting a subset of chunks discards rather than concentrates relevant information. Additionally, the single-query retrieval strategy with the first 300 characters of the note as a query is poorly suited to capture the full scope of a BHC, particularly under the Flan-T5-Large 512-token budget. Together with the VeriFact-BHC results, this ablation supports a unifying interpretation: *RAG confers value precisely in proportion to the degree to which the source documentation is fragmented and exceeds the model's context window.* This ablation also differs from the primary experiment in model architecture (encoder-decoder vs. decoder-only); fully disentangling the effects of model capacity and input structure remains a direction for follow-on work.

## 6 Discussion

**No single metric captures BHC quality.** Our reference-based metrics (ROUGE, BERTScore) and proposition-level recall converge on the conclusion that zero-shot and budget-matched RAG produce summaries of comparable quality. The LLM-as-judge analysis discriminates more finely on completeness, conciseness, and overall quality, but compresses across factuality and hallucination. The qualitative anchor case (Section 5.5) shows that clinically significant differences—a fabricated admission diagnosis under baseline that RAG corrects—can exist without any of these metrics flagging them. The methodological lesson is that triangulating across automated metrics, structured-attribute scoring, and targeted qualitative review yields a more honest picture of summarization quality than any one method alone, an observation aligned with the LLM-as-judge literature [6, 7] and with the VeriFact authors' motivation for proposition-level evaluation [3].

**RAG's primary win at this scale is efficiency, not accuracy.** On the VeriFact-BHC cohort, RAG v3 matches baseline on every aggregate automated metric while consuming roughly 42% less input context on average. For both model families, the cross-condition gap (zero-shot vs. RAG v3) is smaller than the cross-model gap (Mistral vs. Llama). For deployment-oriented questions—inference cost per patient, latency, GPU memory—this is a meaningful finding: the same-quality summary at substantially lower compute is the actionable result, even in the absence of an accuracy improvement.

**Cross-model robustness.** The $2\times2$ design lets us check whether findings hold across model families. They do, on the metrics evaluated: both Mistral and Llama show flat zero-shot-vs-RAG profiles on ROUGE-L and BERTScore, and both show the same truncation-subgroup pattern (worse scores under truncation). Llama scores slightly below Mistral on both metrics, but the cross-model gap is consistent across both input regimes; we have no evidence that one model family benefits more from RAG than the other at this scale.

**When does RAG matter?** The MIMIC-IV-Ext-BHC ablation provides the clearest single piece of methodological guidance from this study. On the curated single-document dataset, RAG underperforms baseline on every metric, while on the fragmented multi-source dataset (VeriFact-BHC), RAG matches baseline on aggregate metrics and modestly improves on judge-rated completeness. We read this as a structural claim: RAG is most valuable when the source documentation is fragmented and exceeds the context window, and it can actively hurt when the source is already compact and curated. Designing for retrieval makes sense in the multi-source ICU setting; it does not generalize to settings where the retrieval step has nothing useful to do.

**Limitations.** Several limitations qualify our findings. First, the LLM-as-judge analysis used a custom five-dimension rubric rather than the validated PDSQI-9 instrument, and was not calibrated against physician scoring on a reliability sub-sample. The compressed hallucination distribution we observed is consistent both with the calibration prompt explicitly anchoring scores to the middle of the 1–5 range and with limitations of a single judge model at this output length; we cannot distinguish these without physician comparison. Second, the judge was applied to Mistral conditions only; extending it to the Llama conditions is a near-term extension. Third, MIMIC-III data are vintage 2001–2012 and may not reflect contemporary documentation styles; findings should be interpreted as applying to that documentation regime rather than to current EHR practice without further validation. Fourth, the cohort is single-site (Beth Israel Deaconess Medical Center); cross-site generalization is not tested at this scale. Fifth, the proposition-recall metric uses semantic similarity at a fixed cosine threshold (0.60) rather than entailment-based verification, and is therefore more permissive than a fact-checking judge would be. Finally, the MIMIC-IV ablation differs from the primary experiment in both dataset and model architecture; isolating the contribution of input structure from model capacity requires additional matched experiments.

## 7 Conclusion and Future Work

We presented a multi-model, multi-evaluator study of LLM-based summarization of fragmented ICU clinical notes. Across 100 MIMIC-III patients, two open-source generation models (Mistral 7B Instruct, Llama 3.1 8B Instruct), and three complementary evaluation methods (reference-based metrics, proposition-level fact recall, LLM-as-judge), the consistent finding is that budget-matched retrieval-augmented generation achieves quality comparable to naive zero-shot prompting while consuming roughly 42% less input context. The LLM-as-judge analysis identifies modest but systematic improvements for RAG on completeness and overall quality, while a cross-dataset ablation on the curated MIMIC-IV-Ext-BHC corpus shows that RAG underperforms baseline when the source documentation does not require it. A qualitative anchor case demonstrates that clinically meaningful hallucination differences can persist invisibly to all of the automated metrics we evaluated, motivating future evaluation work that combines multiple automated signals with targeted clinical review.

Several directions follow naturally from this work. **(i) Physician-calibrated judge evaluation.** Applying the validated PDSQI-9 instrument with a physician-calibrated LLM-as-judge configuration on a stratified subset would provide a reliability anchor for the exploratory judge results presented here. **(ii) Symmetric judge coverage.** Extending the judge analysis to the Llama conditions would close the asymmetry in our $2 \times 2$ multi-evaluator design. **(iii) Comprehensive case-presentation summaries.** A natural extension of BHC generation is a structured ten-section patient snapshot that mirrors how residents present patients on rounds, which would test whether the structure of the output (free narrative vs. structured fields) affects clinical utility independently of the underlying retrieval and model choices. **(iv) Cross-site validation.** Replicating the full pipeline on a second institution's notes would test the generalizability of the findings beyond a single hospital's documentation conventions.

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